Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: [TEAM NAME]
b. Team leader name: Khunapat Prapassornchaikul
c. Problem Statement: Media, Content & Digital Experiences. FaceFinder turns one shared Google Drive event album into one folder per person, with Gemini picking and sorting the best photos.

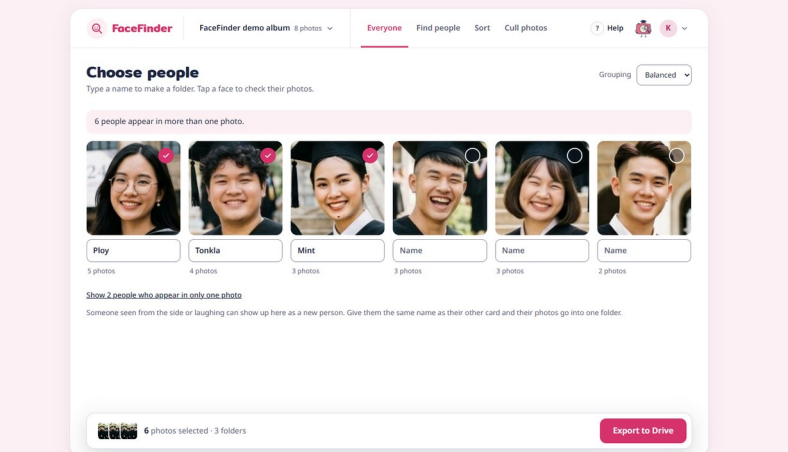
Brief about the idea

The problem

After a graduation, wedding or company event, the photographer shares one Google Drive folder with hundreds of photos. Finding yourself, or sending each friend their photos, means an hour of scrolling, and the photographer still culls blurry and eyes-closed shots by hand.

FaceFinder

Paste the folder link and FaceFinder scans the album once. A face-recognition model groups every face into people; Gemini on Vertex AI judges each photo and describes it. Anyone can then find a person, keep the best shots, sort by plain-word descriptions, and copy one folder per person into their own Drive. The original album is never changed.



The live app on the demo album. The people are made up, created with Gemini.

Opportunities

How is it different?

Google Photos groups faces only inside one person's own library. Event photos usually live in one shared Drive folder, with no face search and no culling. FaceFinder works on any shared Drive folder, for everyone who can open it, and gives back real Drive folders.

How does it solve the problem?

One scan per album. Then name a face or upload one selfie to get that person's photos. Gemini suggests Keep, Review or Rejected with a reason for each photo, and sorts the album by a plain-word description. One click copies it all into folders in your Drive.

USP

Two AIs, each doing what it does best: a face model decides who is in a photo, Gemini decides how good it is and what is in it. Copy-only and privacy-first: no photos stored, data deletable at any time. Works on a phone, nothing to install.

List of features offered by the solution



Everyone

Every face grouped into people. Type a name to make that person's folder.



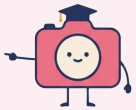
Find people + Best 10

One selfie finds a person. Best 10 picks their best shots, each with Gemini's reason.



Cull photos

Keep, Review or Rejected, with reasons: blur, eyes closed, weaker shots of a burst.



Sort by description

Describe a folder in plain words ("throwing caps"). Gemini fills it and suggests folders.



Export to Drive

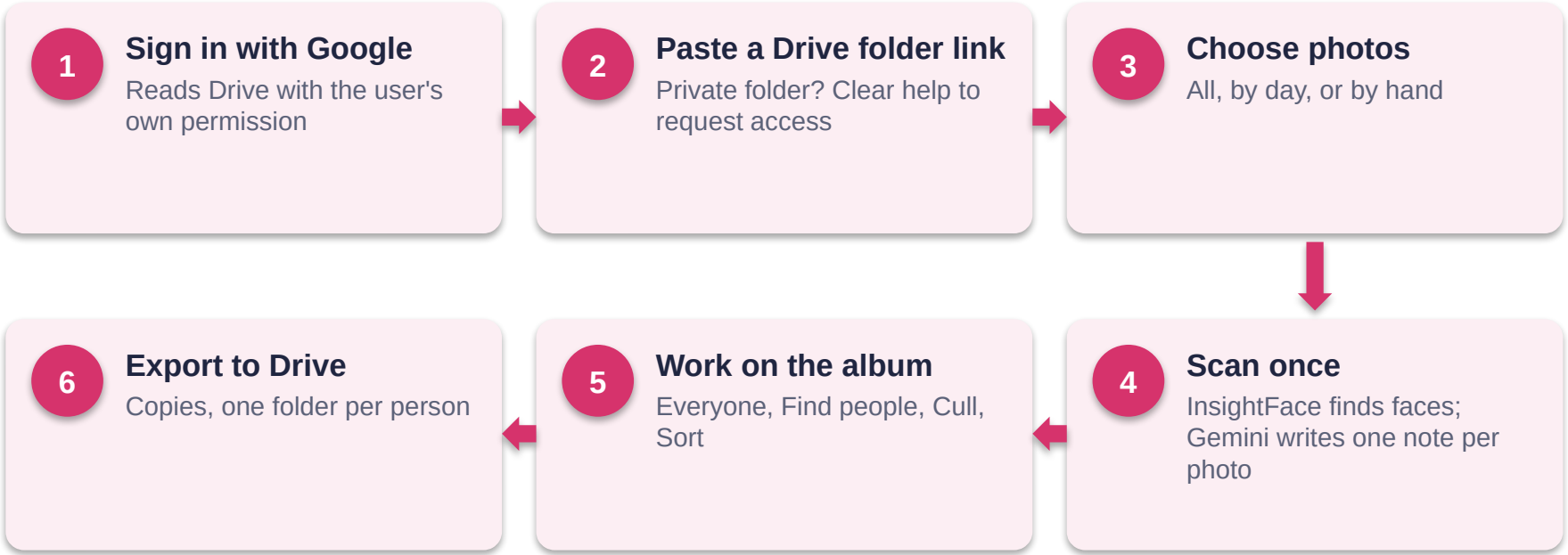
One folder per person in your own Drive. Copies are made inside Google; never duplicated.



Control and privacy

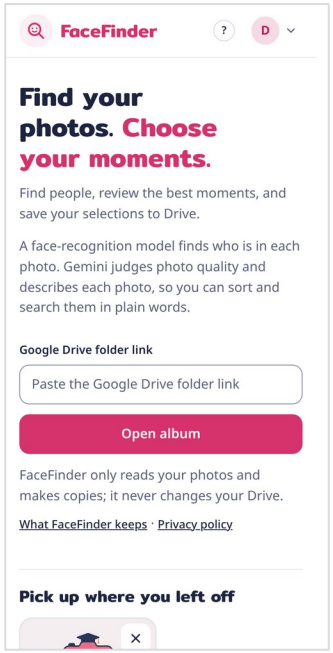
Choose photos by day or by hand, stop and resume. See what is kept, delete it any time.

Process flow diagram or Use-case diagram

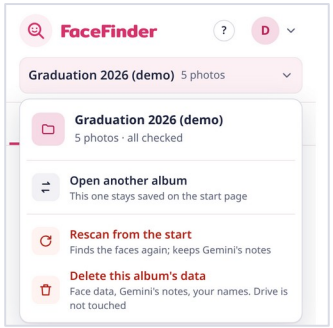


Who uses it: the photographer culls and sorts; guests find themselves; friends share each person's folder.

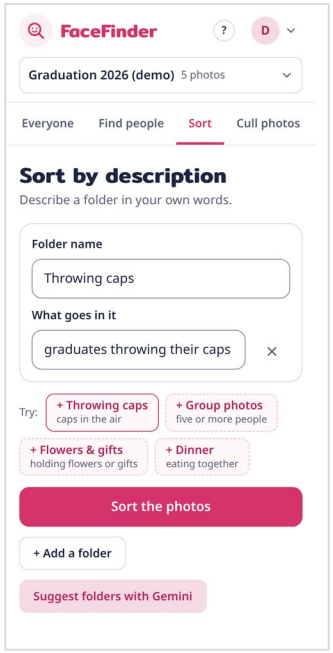
Wireframes/Mock diagrams of the proposed solution (optional)



Start page



Album menu

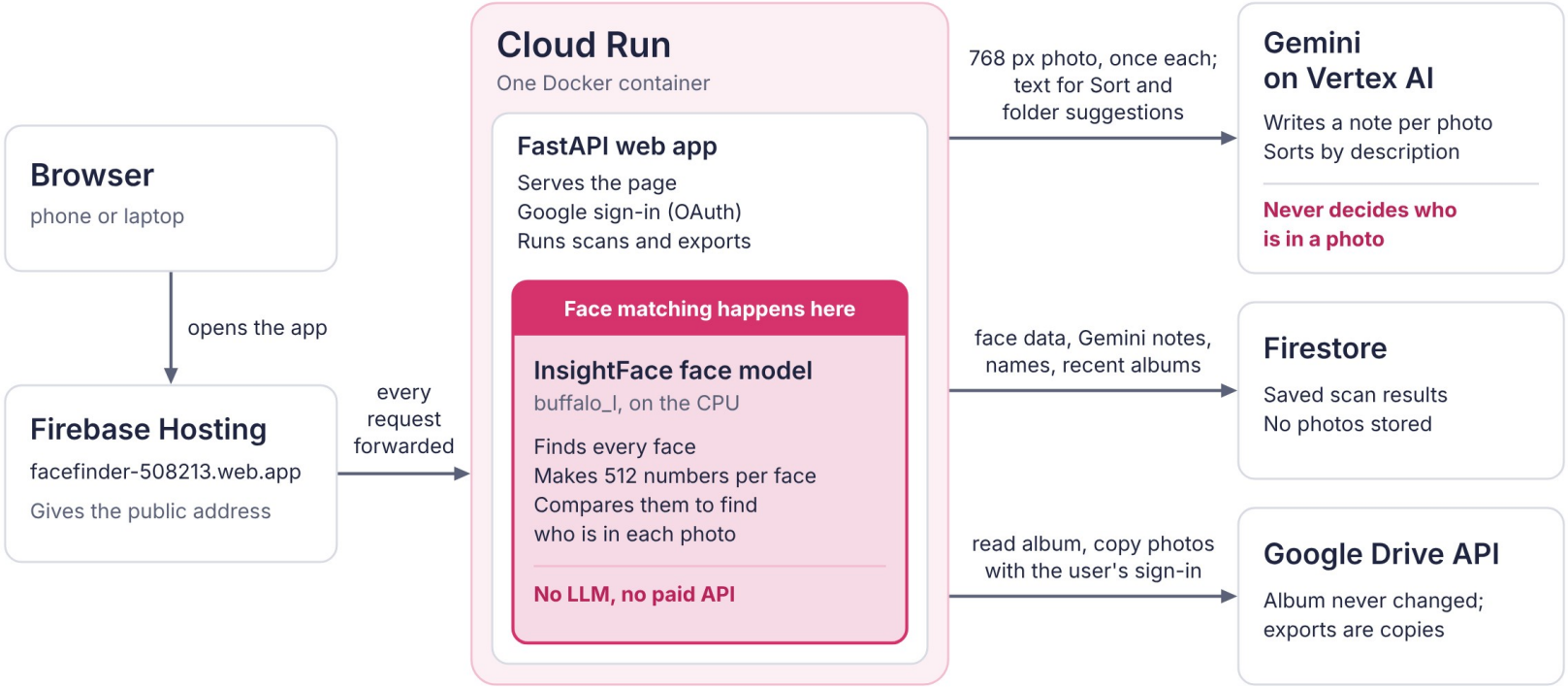


Sort by description

Phone first

Works from 320 px wide to desktop, with large touch targets and a bottom action bar. Our own pink design and camera mascot.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Google Cloud

Cloud Run: runs the app and the face model

Firestore: scan results, Gemini notes, names

Firebase Hosting: the public address

Vertex AI: Gemini, no API key needed

Drive API, Google sign-in: read albums, copy photos

Secret Manager: sign-in secrets

AI models

Gemini 3.5 Flash: photo notes, sorting, folder suggestions, structured JSON

InsightFace buffalo_l: face detection and 512-number face matching, ONNX Runtime on CPU

OpenCV: image handling, burst fingerprints

App and tools

Python, FastAPI: server and background jobs

HTML, CSS, JavaScript: one-file front end

Docker: one container

Tests: 90 Python checks, Playwright browser tests

Antigravity, Gemini: browser testing, demo photos of made-up people

Estimated implementation cost (optional)

What it costs to run

- **Cloud Run:** one 4-vCPU instance that scales to zero, so an idle app has no compute cost.
- **Gemini:** one call per photo, once, on a 768 px copy. Results are cached, so rescans cost nothing. Sort sends text only, 40 photos per call.
- **Free after the scan:** Find people, Cull and Best 10 reuse the saved notes; face matching runs in our container, with no per-photo fee.
- **Firestore and Drive:** one document per photo; copies are made inside Google, with no download or upload.

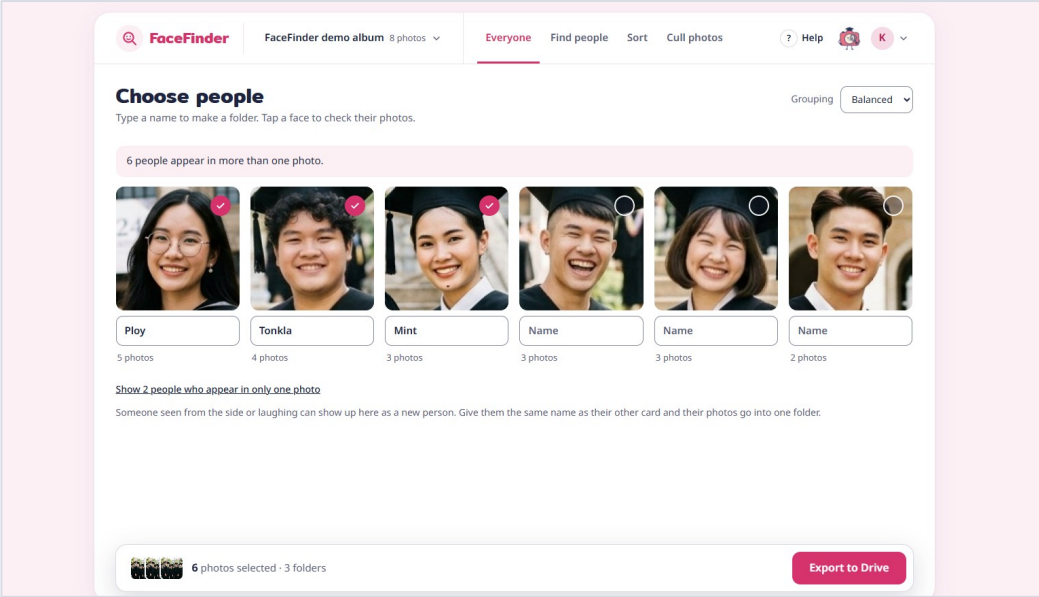
Per 1,000 photos

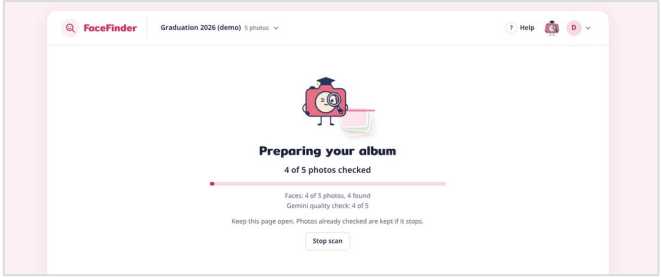
1,000
Gemini image calls, once

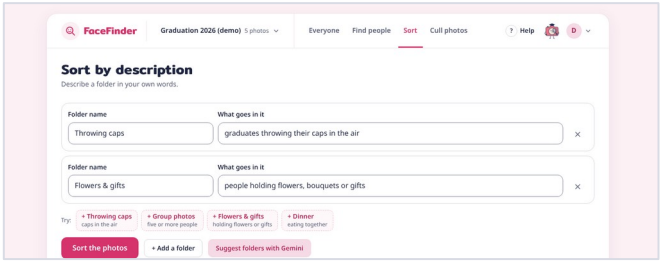
~25
text calls for each sort

0
Gemini calls to rescan

Snapshots of the prototype







Everyone on the live app: name a face to make a folder. Right: scanning, and Sort by description.

Prototype Performance report/Benchmarking

0.75–0.98
face match score for the same person, against −0.05 for a different person

40 of 40
faces 18 px wide (the back rows of a class photo) found, using two detection sizes

~2.6 s
per photo for a live scan with Gemini notes (37 photos in 1 min 36 s, 4 vCPU)

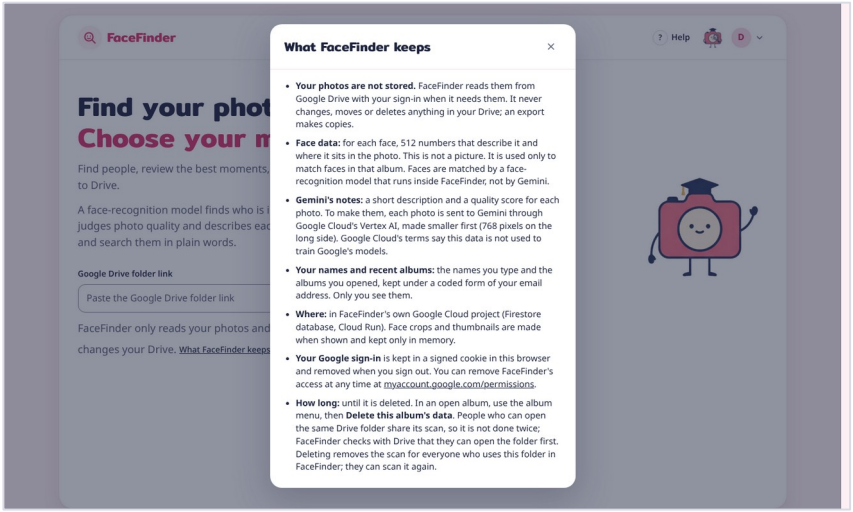
45 s
to export 54 copies into 3 person folders in Google Drive

~1 s
to find a person from one selfie once the album is scanned

~5 s
for Stop to end a scan; a rescan makes 0 new Gemini calls

Measured on the live app by test users on 17–18 October 2026, and by the repository's automatic tests.

Additional Details/Future Development (if any)



Trust today: no photos stored, a plain list of what is kept, public privacy policy, and "Delete this album's data" in the album menu.

Next steps

- **Scale:** move scans and exports to Cloud Run jobs or Cloud Tasks; Firestore already holds all state.
- **Guest mode:** a QR code at the event so each guest finds their own photos.
- **Commercial release:** a commercially licensed face model (the InsightFace model files are for non-commercial use).
- **Google verification,** so the "unverified app" warning goes away.
- **More sources:** Google Photos shared albums, and video.

Provide links to your:

- GitHub Public Repository: https://github.com/Khunapat/facefinder
- Demo Video Link (3 to 4 minutes): [ADD VIDEO LINK]
- Final Product Link: https://facefinder-508213.web.app

For judges

Open the app and sign in with Google. Google may show "Google hasn't verified this app": press Advanced, then Go to facefinder. Open the Sample album card (made-up people created with Gemini) and try Everyone, Find people, Sort and Cull photos.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

